ggplot2数据可视化::速查表

基础

ggplot2 基于图形语法，使用相同的组件（数据集、坐标系和表示数据点的几何对象）来构建图片。



为了获取显示值，数据中的变量映射到图形的视觉属性，如大小、颜色以及x和y位置。



完成以下模板来构建图形

必要

```
ggplot (data = <DATA> ) +
<GEOM_FUNCTION>(mapping = aes(<MAPPINGS>),
 stat = <STAT> , position = <POSITION> ) +
<COORDINATE_FUNCTION> +
<FACET_FUNCTION> +
<SCALE_FUNCTION> +
<THEME_FUNCTION>
```

非必要，有默认值

ggplot(data = mpg, aes(x = cty, y = hwy)) 通过添加图层来完成图形，每层添加一个geom函数。

last_plot() 返回上一个图片。

ggsave("plot.png", width = 5, height = 5) 将最后一个图片保存至工作目录中名为“plot.png”的5'x 5'文件。 文件类型与文件扩展名相匹配。

参数 通用参数

color and fill - string ("red", "#RRGGBB")

linetype - integer or string (0 = "blank", 1 = "solid", 2 = "dashed", 3 = "dotted", 4 = "dotdash", 5 = "longdash", 6 = "twodash")

lineend - string ("round", "butt", or "square")

linejoin - string ("round", "mitre", or "bevel")

size - integer (line width in mm)

shape - integer/shape name or a single character ("a")



几何对象

使用geom函数表示数据点，使用geom的属性表示变量。 每个函数绘制一个图层。

基本图像

```
a <- ggplot(economics, aes(date, unemploy))
b <- ggplot(seals, aes(x = long, y = lat))
```

a + geom_blank() and **a + expand_limits()**
Ensure limits include values across all plots.

b + geom_curve(aes(yend = lat + 1, xend = long + 1), curvature = 1) - x, xend, y, yend, alpha, angle, color, curvature, linetype, size

a + geom_path(lineend = "butt", linejoin = "round", linemitre = 1)
x, y, alpha, color, group, linetype, size

a + geom_polygon(aes(alpha = 50)) - x, y, alpha, color, fill, group, subgroup, linetype, size

b + geom_rect(aes(xmin = long, ymin = lat, xmax = long + 1, ymax = lat + 1)) - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

a + geom_ribbon(aes(ymin = unemploy - 900, ymax = unemploy + 900)) - x, ymax, ymin, alpha, color, fill, group, linetype, size

分段线

常用参数: x, y, alpha, color, linetype, size

b + geom_abline(aes(intercept = 0, slope = 1))
b + geom_hline(aes(yintercept = lat))
b + geom_vline(aes(xintercept = long))

b + geom_segment(aes(yend = lat + 1, xend = long + 1))
b + geom_spoke(aes(angle = 1:1155, radius = 1))

单变量 连续

```
c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)
```

c + geom_area(stat = "bin")
x, y, alpha, color, fill, linetype, size

c + geom_density(kernel = "gaussian")
x, y, alpha, color, fill, group, linetype, size, weight

c + geom_dotplot()
x, y, alpha, color, fill

c + geom_freqpoly()
x, y, alpha, color, group, linetype, size

c + geom_histogram(binwidth = 5)
x, y, alpha, color, fill, linetype, size, weight

c2 + geom_qq(aes(sample = hwy))
x, y, alpha, color, fill, linetype, size, weight

离散

```
d <- ggplot(mpg, aes(fl))
```

d + geom_bar()
x, alpha, color, fill, linetype, size, weight

双变量

两个都连续

```
e <- ggplot(mpg, aes(cty, hwy))
```

e + geom_label(aes(label = cty), nudge_x = 1, nudge_y = 1) - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

e + geom_point()
x, y, alpha, color, fill, shape, size, stroke

e + geom_quantile()
x, y, alpha, color, group, linetype, size, weight

e + geom_rug(sides = “bl")
x, y, alpha, color, linetype, size

e + geom_smooth(method = lm)
x, y, alpha, color, fill, group, linetype, size, weight

e + geom_text(aes(label = cty), nudge_x = 1, nudge_y = 1) - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

一个离散，一个连续

```
f <- ggplot(mpg, aes(class, hwy))
```

f + geom_col()
x, y, alpha, color, fill, group, linetype, size

f + geom_boxplot()
x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

f + geom_dotplot(binaxis = "y", stackdir = “center") x, y, alpha, color, fill, group

f + geom_violin(scale = “area")
x, y, alpha, color, fill, group, linetype, size, weight

两个都离散

```
g <- ggplot(diamonds, aes(cut, color))
```

g + geom_count()
x, y, alpha, color, fill, shape, size, stroke

e + geom_jitter(height = 2, width = 2)
x, y, alpha, color, fill, shape, size

三变量

```
seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)); l <- ggplot(seals, aes(long, lat))
```

l + geom_contour(aes(z = z))
x, y, z, alpha, color, group, linetype, size, weight

l + geom_contour_filled(aes(fill = z))
x, y, alpha, color, fill, group, linetype, size, subgroup

l + geom_raster(aes(fill = z), hjust = 0.5, vjust = 0.5, interpolate = FALSE)
x, y, alpha, fill

l + geom_tile(aes(fill = z))
x, y, alpha, color, fill, linetype, size, width

连续二元分布

```
h <- ggplot(diamonds, aes(carat, price))
```

h + geom_bin2d(binwidth = c(0.25, 500))
x, y, alpha, color, fill, linetype, size, weight

h + geom_density_2d()
x, y, alpha, color, group, linetype, size

h + geom_hex()
x, y, alpha, color, fill, size

连续函数

```
i <- ggplot(economics, aes(date, unemploy))
```

i + geom_area()
x, y, alpha, color, fill, linetype, size

i + geom_line()
x, y, alpha, color, group, linetype, size

i + geom_step(direction = "hv")
x, y, alpha, color, group, linetype, size

可视化误差

```
df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)
j <- ggplot(df, aes(grp, fit, ymin = fit - se, ymax = fit + se))
```

j + geom_crossbar(fatten = 2) - x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + geom_errorbar() - x, ymax, ymin, alpha, color, group, linetype, size, width
Also **geom_errorbarh()**.

j + geom_linerange()
x, ymin, ymax, alpha, color, group, linetype, size

j + geom_pointrange() - x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

地图

```
data <- data.frame(murder = USArrests$Murder,
                   state = tolower(rownames(USArrests)))
map <- map_data("state")
k <- ggplot(data, aes(fill = murder))
```

k + geom_map(aes(map_id = state), map = map) **+ expand_limits**(x = map$long, y = map$lat)
map_id, alpha, color, fill, linetype, size

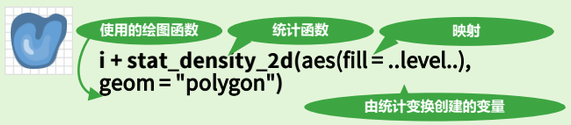
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统计变换 另一种构建图层的方法

统计变换构建新变量来绘图（例如，count，prop）。



通过更改geom函数的默认统计信息，**geom_bar(stat="count")** 或者使用统计变化功能来绘图**stat_count(geom="bar")**，其调用默认图片来创建一个图层（相当于geom函数）。
使用 **..name..** 语法将统计变化映射到坐标。



c + **stat_bin**(binwidth = 1, boundary = 10)
x, y | ..count.., ..ncount.., ..density.., ..ndensity..

c + **stat_count**(width = 1) x, y | ..count.., ..prop..

c + **stat_density**(adjust = 1, kernel = "gaussian")
x, y | ..count.., ..density.., ..scaled..

e + **stat_bin_2d**(bins = 30, drop = T)
x, y, fill | ..count.., ..density..

e + **stat_bin_hex**(bins = 30) x, y, fill | ..count.., ..density..

e + **stat_density_2d**(contour = TRUE, n = 100)
x, y, color, size | ..level..

e + **stat_ellipse**(level = 0.95, segments = 51, type = "t")

l + **stat_contour**(aes(z = z)) x, y, z, order | ..level..

l + **stat_summary_hex**(aes(z = z), bins = 30, fun = max)
x, y, z, fill | ..value..

l + **stat_summary_2d**(aes(z = z), bins = 30, fun = mean)
x, y, z, fill | ..value..

f + **stat_boxplot**(coef = 1.5)
x, y | ..lower.., ..middle.., ..upper.., ..width.. , ..ymin.., ..ymax..

f + **stat_ydensity**(kernel = "gaussian", scale = "area") x, y |
..density.., ..scaled.., ..count.., ..n.., ..violinwidth.., ..width..

e + **stat_ecdf**(n = 40) x, y | ..x.., ..y..

e + **stat_quantile**(quantiles = c(0.1, 0.9),
formula = y ~ log(x), method = "rq") x, y | ..quantile..

e + **stat_smooth**(method = "lm", formula = y ~ x, se = T,
level = 0.95) x, y | ..se.., ..x.., ..y.., ..ymin.., ..ymax..

ggplot() + xlim(-5, 5) + **stat_function**(fun = dnorm,
n = 20, geom = "point") x | ..x.., ..y..

ggplot() + **stat_qq**(aes(sample = 1:100))
x, y, sample | ..sample.., ..theoretical..

e + **stat_sum**() x, y, size | ..n.., ..prop..

e + **stat_summary**(fun.data = "mean_cl_boot")

h + **stat_summary_bin**(fun = "mean", geom = "bar")

e + **stat_identity**()

e + **stat_unique**()

标尺 使用scales包覆盖默认值

将映射数据缩放到较为美观的比例。
添加新的标尺来改变映射。

n <- d + geom_bar(aes(fill = fl))



标尺的一般用法

使用大多数参数
scale_*_continuous() - 将数据的连续取值映射为图形属性的取值
scale_*_discrete() - 将数据的离散取值映射为图形属性的取值
scale_*_binned() - 将数据的连续取值映射为离散的统计堆
scale_*_identity() - 使用数据的值作为图形属性的取值
scale_*_manual(values = c()) - 将数据的离散取值作为手工指定的图形属性的取值
scale_*_date(date_labels = "%m/%d"),
date_breaks = "2 weeks") - 将数据值视为日期
scale_*_datetime() - 将数据x视为时间
参数和scale_x_date()一样。有关标签格式请参阅?striptime。

调整X和Y的比例

调整x和y的标尺(使用x为例)
scale_x_log10() - 以log10比例绘制x
scale_x_reverse() - 反转x轴方向
scale_x_sqrt() - 以平方根绘制x

颜色和填充比例（离散）

n + **scale_fill_brewer**(palette = "Blues")
选择调色板: RColorBrewer::display.brewer.all()
n + **scale_fill_grey**(start = 0.2, end = 0.8,
na.value = "red")

颜色和填充比例（连续）

o <- c + geom_dotplot(aes(fill = ..x..))
o + **scale_fill_distiller**(palette = "Blues")

o + **scale_fill_gradient**(low="red", high="yellow")

o + **scale_fill_gradient2**(low = "red", high = "blue", mid = "white", midpoint = 25)

o + **scale_fill_gradientn**(colors = topo.colors(6))
也见: rainbow(), heat.colors(), terrain.colors(), cm.colors(), RColorBrewer::brewer.pal()

形状和尺寸比例

p <- e + geom_point(aes(shape = fl, size = cyl))
p + **scale_shape**() + **scale_size**()
p + **scale_shape_manual**(values = c(3:7))

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

p + **scale_radius**(range = c(1,6))
p + **scale_size_area**(max_size = 6)

坐标系

r <- d + geom_bar()

r + **coord_cartesian**(xlim = c(0, 5)) - xlim, ylim
默认笛卡尔坐标系

r + **coord_fixed**(ratio = 1/2)
ratio, xlim, ylim - x和y单位之间固定长宽比的笛卡尔坐标

ggplot(mpg, aes(y = fl)) + **geom_bar**()
通过切换x和y参数映射翻转笛卡尔坐标

r + **coord_polar**(theta = "x", direction=1)
theta, start, direction – 极坐标

r + **coord_trans**(y = "sqrt") - x, y, xlim, ylim
转换后的笛卡尔坐标。将xtrans和ytrans设置为窗口函数的名称。

π + **coord_quickmap**()

π + **coord_map**(projection = "ortho",
orientation = c(41, -74, 0)) - projection, xlim, ylim
从mapproj包中映射投影(mercator (default), azequalarea, lagrange, etc.)

位置调整

位置调整决定了如何安排原本会占据相同空间的图例

s <- ggplot(mpg, aes(fl, fill = drv))

s + **geom_bar**(position = "dodge")
并排排列元素

s + **geom_bar**(position = "fill")
堆叠元素并标准化高度

e + **geom_point**(position = "jitter")
将随机抖动添加到每个元素的X和Y位置以避免重叠

e + **geom_label**(position = "nudge")
标签稍远离数据点

s + **geom_bar**(position = "stack")
堆叠元素

每个位置调整都可以重新编写为具有手动宽度和高度参数的函数:
s + geom_bar(position = position_dodge(width = 1))

主题

r + **theme_bw**()
网格白色背景

r + **theme_gray**()
灰色背景（默认主题）

r + **theme_dark**()
黑色背景

r + **theme_classic**()

r + **theme_light**()

r + **theme_linedraw**()

r + **theme_minimal**()
简单主题

r + **theme_void**()
清空主题

r + **theme**() 自定义主题的各个方面，例如轴、图例、面板和构面属性。
r + ggtitle("Title") + theme(plot.title.postion = "plot")
r + theme(panel.background = element_rect(fill = "blue"))

分面

根据一个或多个离散变量划分子图。

t <- ggplot(mpg, aes(cty, hwy)) + geom_point()

t + **facet_grid**(cols = vars(fl))
基于fl的列分面

t + **facet_grid**(rows = vars(year))
基于year的行分面

t + **facet_grid**(rows = vars(year), cols = vars(fl))
列和行的分面图

t + **facet_wrap**(vars(fl))
包裹成矩形布局的分面图

设置scales限制分面坐标轴
t + **facet_grid**(rows = vars(drv), cols = vars(fl),
scales = "free")
x和y轴适应各自的分面
"free_x" - 限制调整x轴
"free_y" - 限制调整y轴

设置labeller属性调整分面的标签
t + **facet_grid**(cols = vars(fl), labeller = label_both)
t + **facet_grid**(rows = vars(fl),
labeller = label_bquote(alpha ^ .(fl)))

fl: c	fl: d	fl: e	fl: p	fl: r
α^c	α^d	α^e	α^p	α^r

标签与图例

使用labs()标记图中的元素。
t + **labs**(x = "New x axis label", y = "New y axis label",
title ="Add a title above the plot",
subtitle = "Add a subtitle below title",
caption = "Add a caption below plot",
alt = "Add alt text to the plot",
<AES> = "New <AES> legend title")

t + **annotate**(geom = "text", x = 8, y = 9, label = "A")
Places a geom with manually selected aesthetics.使用手动选择的参数调整几何对象的放置位置。

p + **guides**(x = guide_axis(n.dodge = 2)) 使用guide_axis(n.dodge或 angle)避免拥挤或重叠的标签。

n + **guides**(fill = "none")设置图例类型: colorbar, legend, or none (no legend)

n + **theme**(legend.position = "bottom")
放置图例: "bottom", "top", "left", or "right"

n + **scale_fill_discrete**(name = "Title",
labels = c("A", "B", "C", "D", "E"))
使用scale函数设置图例标签

缩放



没有裁剪（推荐）

t + **coord_cartesian**(xlim = c(0, 100), ylim = c(10, 20))



裁剪（删除看不见的数据点）

t + **xlim**(0, 100) + **ylim**(10, 20)

t + **scale_x_continuous**(limits = c(0, 100)) +
scale_y_continuous(limits = c(0, 100))

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